

COMPLEXITY AND RECURRENCES

Algorithmic Performance and Mathematical Analysis

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Course Outline

1 Introduction to Complexity Analysis

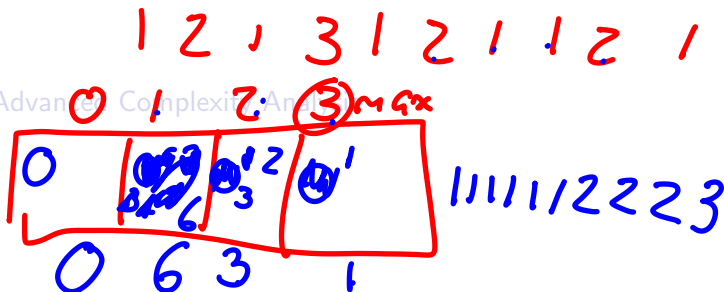
2 Advanced Complexity Analysis



Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis



Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, which one should we choose?

Counting Sort $\rightarrow$ repetitions

① Max $\rightarrow$ number of list

②  $\rightarrow$ $len = max + 1$

③ for ∇ in numbers:
 $result[i] += 1$

④ print $result[i] \rightarrow i$

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Given multiple algorithms that solve the same problem, **which one should we choose?**

- **Performance:** How fast does the algorithm run?
- **Resources:** How much memory does it use?
- **Scalability:** How does it handle large inputs?
- **Predictability:** Can we predict its behavior?

contest

Real World Impact

Condor

- User experience depends on algorithm speed
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions



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Real-World Impact

- User experience depends on algorithm **speed**
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions

\$/hour

Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

cost } → machine

Arithmetic: $+$, $-$, $*$, $/$

$=$, $<$, $>$

array[i]

Simple built-in functions



Elementary Operations Identification I

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Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
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- **Function Calls:** Simple built-in functions

no relevant

1 step

sum functions step



Elementary Operations Identification II

Example

Count operations in simple code:

```

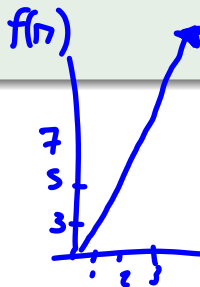
1: sum = 0 // 1 operation
2: for i = 0 to n - 1 do
3:   sum = sum + array[i] // 2 operations per iteration
4: end for
  
```

Handwritten annotations: A red arrow points from the '1' in line 1 to a circled 'n' with the word 'times' next to it. A red bracket underlines the 'for' loop in line 2. A pink bracket underlines the assignment in line 3, with an arrow pointing to the '2' in '2 operations per iteration'.

Total: 1 + 2n operations

Handwritten annotation: A red circle around '2n' with an arrow pointing to the text 'growth factor'.

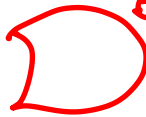
growth factor } loop



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

- 1: Copy all elements from A to C
 - 2: **for** each element ~~in A~~  + 
 - 3: Check if element is in C
 - 4: **if** element not found **then**
 - 5: Add element to C
 - 6: **end if**
 - 7: **end for**
- $A \cup (B - A)$



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

- 1: Copy all elements from A to C
- 2: **for** each element in B **do**
- 3: Check if element exists in C
- 4: **if** element not found **then**
- 5: Add element to C
- 6: **end if**
- 7: **end for**

$\sim$ each element (n)

$\sim$ n times

Case $A \cap B = \emptyset \rightarrow n + n (2)$

Case $A = B \rightarrow n + n (1)$

Java-loop
Python: `in`



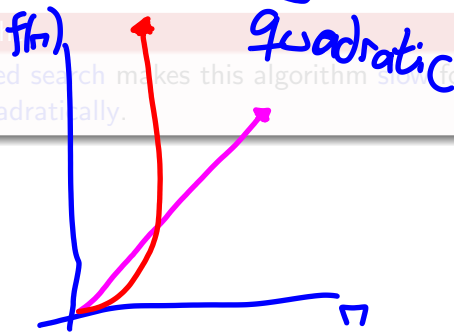
Study Case: Counting Operations in Arrays Union II

Operation Analysis

- Copy from A: n operations *length A*
- For each element in B: m iterations *length B*
- Search in C: Up to n comparisons per element
- Total worst case: $n + m * n = n(1 + m)$ operations

Key Insight

The nested search makes this algorithm slow for large arrays! Performance grows quadratically.



- **Copy from A:** n operations
- **For each element in B:** m iterations
- **Search in C:** Up to n comparisons per element
- **Total worst case:** $n + m * n = n(1 + m)$ operations



The nested search makes this algorithm slow for large arrays. Performance grows quadratically.

$$\left. \begin{array}{l} \text{for } \dots \\ \text{for } \dots \end{array} \right\} \begin{array}{l} n \approx m \\ \approx n \times m^2 \end{array}$$

for for for } 3



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations



Average Case

- Expected performance
- Typical input
- Example: Finding element in middle



Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found

$$I \leq n \leq 1000$$

Context evaluation



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

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- Typical input
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- Example: Finding element at first position

Average Case

- Expected performance
 - Typical input
 - Example: Finding element in middle
- Handwritten notes: A red bracket groups the first two items, with "C?" next to it. A red arrow points from the word "Linear" (written below) to the "Example" item.*

Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

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- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Worst Case

- Maximum operations
- Least favorable input
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Best, Average, Worst Case Scenarios II

Example

Search in array [5, 2, 8, 1, 9]:

- **Best:** Search for 5 $\rightarrow$ 1 comparison
- **Average:** Random element $\rightarrow \sim 2.5$ comparisons
- **Worst:** Search for 7 (not found) $\rightarrow$ 5 comparisons

Linear



Performance Prediction Methods

Approaches to Predict Algorithm Performance

① Theoretical Analysis: Mathematical analysis

② Empirical Analysis: Timing actual executions

③ Simulation: Testing with synthetic data

④ Profiling: Detailed runtime analysis

for $\rightarrow$ 7 times

$x = y * z \rightarrow$ 1 step $*$
1 step $=$ 2 steps



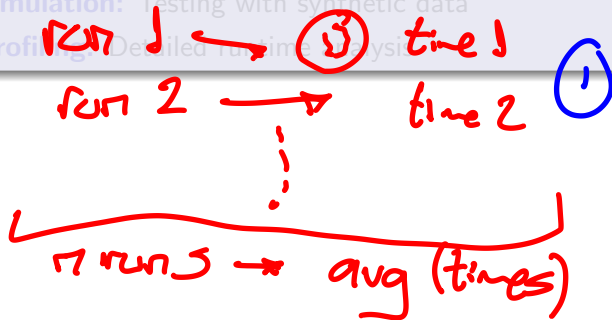
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Performance Prediction Methods

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- 1 **Theoretical Analysis:** Mathematical analysis
- 2 **Empirical Analysis:** Timing actual executions
- 3 **Simulation:** Testing with synthetic data
- 4 **Profiling:** Detailed runtime analysis

random
array

fake
data



Performance Prediction Methods

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- 1 **Theoretical Analysis:** Mathematical analysis
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- 4 **Profiling:** Detailed runtime analysis

static

run

memory
time
network



Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual complexity
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input coverage
- Time-consuming

10 20

2.7

7²

best
avg
worst

~1000000
~100.000.000

Best Practice

Use theoretical analysis for algorithm design, then validate with empirical testing.

Performance Prediction Methods II

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Time vs. Space Trade-offs I

Definition

A time-space trade-off occurs when we can reduce time by using more memory, or reduce memory at the cost of increased time.

Trade-off Decision Factors:

- Available memory constraints
- Time requirements
- Cost of computation vs. storage
- Frequency of algorithm execution

Bank search



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Cache



more memory
less time



Time vs. Space Trade-offs II

Example

Fibonacci Calculation:

Space-Efficient (Recursive): ✓

- Uses minimal extra memory
- But recalculates same values repeatedly
- Very slow for large numbers

Time-Efficient (Memoization): ✓

- Stores previously calculated values
- Uses more memory
- Much faster for large numbers



var1 var2 var3



Systematic Complexity Calculation Methods I

Step-by-Step Approach

- ① **Identify:** All loops and recursive calls
- ② **Count:** Operations in each section
- ③ **Multiply:** Operations by execution frequency
- ④ **Sum:** All contributions
- ⑤ **Simplify:** Focus on dominant terms



Systematic Complexity Calculation Methods II

Example

Nested Loop Analysis:

```

1: for  $i = 0$  to  $n - 1$  do
2:   for  $j = 0$  to  $n - 1$  do
3:      $result = array[i] + array[j]$ 
4:   end for
5: end for
  
```

Operation Count:

- Outer loop runs n times
 - Inner loop runs n times for each outer iteration
 - Total operations: $n * n = n^2$ operations
-



Study Case: Complexity Calculation for QuickSort I

QuickSort Algorithm Structure

- 1: Choose a **pivot** element
- 2: **Partition** array around pivot
- 3: Recursively sort **left** part
- 4: Recursively sort **right** part

$$T(n) = T(n-1) + T(n-1) + O(n)$$

$$T(n) = T(n-1) + T(n-1) + O(n)$$

Partition Operation

- Compare each element with pivot
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements



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4r

Partition Operation

- Compare each element with pivot
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements

first random



Study Case: Complexity Calculation for QuickSort II

 $M - m \rightarrow \text{pivot}$ (circled)

Best Case

- Pivot divides array in half
- Balanced partitions
- Fast performance



Worst Case

- Pivot is always smallest/largest
- Unbalanced partitions
- Slow performance



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Big O Notation: Definition and Meaning

Definition

Big O notation describes the upper bound of an algorithm's growth rate for large inputs.

What Big O Tells Us:

- How algorithm performance scales with input size
- Worst-case growth rate
- Ignores constant factors and smaller terms
- Focuses on dominant behavior for large inputs



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$$\boxed{\begin{matrix} x^2 + x + 2 \rightarrow x^2 \\ x^3 + 2x^2 + 3x + 1 \rightarrow x^3 \end{matrix}}$$



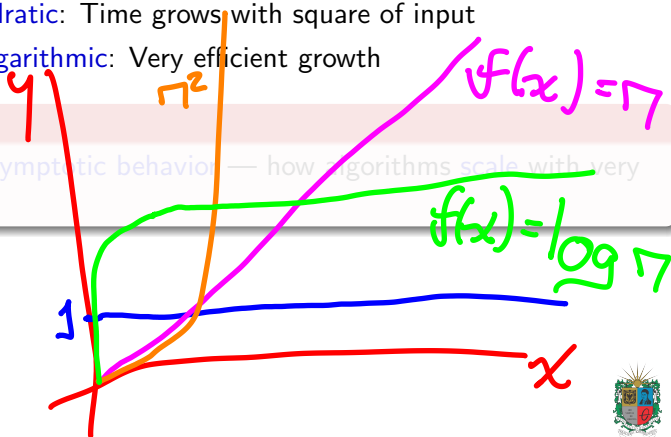
Big O Notation: Classifications

Common Big O Classifications:

- $O(1)$ — **Constant**: Always same time
- $O(n)$ — **Linear**: Time grows with input size
- $O(n^2)$ — **Quadratic**: Time grows with square of input
- $O(\log n)$ — **Logarithmic**: Very efficient growth

Key Insight

Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.



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$n \cdot \log n$ ✓

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Upper Bound Concept and Mathematical Definition

Understanding Upper Bounds

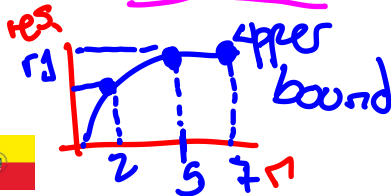
An upper bound guarantees that an algorithm will never perform worse than the bound.

Big O Meaning:

- Sets a ceiling on performance ✓
- Guarantees maximum time
- Used for worst-case analysis
- Helps compare algorithms

Common Growth Orders:

- $O(1)$ — Constant
- $O(\log n)$ — Logarithmic
- $O(n)$ — Linear
- $O(n \log n)$ — Linearithmic
- $O(n^2)$ — Quadratic
- $O(2^n)$ — Exponential



Study Case: First Letter in a Sentence Complexity

Problem

Find the first alphabetical letter in a given sentence string.

```

1: for each character in sentence do
2:   if character is a letter then
3:     return character
4:   end if
5: end for
6: return null
  
```

"hello"

↓ → 7 = len(s)

break

not found

Best: First character is a letter

- Only check 1 character
- Constant time: $O(1)$

Worst: No letters in sentence

- Check every character
- Linear time: $O(n)$

Even though best case is $O(1)$, we report $O(n)$ because Big O represents the worst-case upper bound.

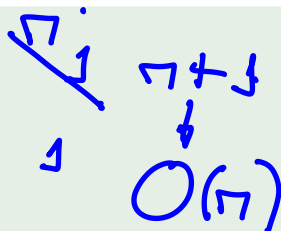
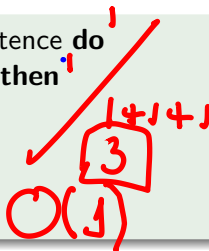
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upper bound

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Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted** array.

```

1: for each element in array do
2:   if element equals target then
3:     return index
4:   end if
5: end for
6: return -1
  
```

- **Best Case:** Element at first position, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element not in array, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in middle, $n/2$ comparisons $\rightarrow O(n)$



Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted array**.

```

1: for each element in array do
2:   if element equals target then
3:     return index
4:   end if
5: end for
6: return -1
  
```

$1 + 1 + 4 \downarrow$
 $3 \Rightarrow O(1)$

$n \downarrow$
 $n + 1 \downarrow$
 $O(n)$

$n/2 \downarrow$
 $n/2 + 1 \downarrow$
 $n/2 + 2 \downarrow$

- **Best Case:** Element at **first position**, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element **not in array**, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in **middle**, $n/2$ comparisons $\rightarrow O(n)$

$O(n)$



Study Case: Binary Search Algorithm Complexity I

Problem

Search for a target element in a **sorted** array.

```

1: while search space not empty do
2:   Find middle element
3:   if middle equals target then
4:     return index
5:   else if middle > target then
6:     search right half
7:   else
8:     search left half
9:   end if
10: end while
11: return -1
  
```

$$\lceil \log_2(n) \rceil$$

$$O(\log n)$$

$$\lceil \log_2(n) \rceil$$

$$\lceil \log_2(n) \rceil$$

$$\lceil n \rceil = \log_e$$

$$a^b = c \Rightarrow \log_a c = b$$



Study Case: Binary Search Algorithm Complexity II

Each step eliminates half of the remaining elements:

- Step 1: n elements
- Step 2: $n/2$ elements
- Step 3: $n/4$ elements
- Step k : $n/2^k$ elements

Algorithm stops when we reach 1 element.

$$\begin{array}{l|l}
 \log n & \log n \\
 2 \log n/2 & \log n/2 \\
 4 \log n/4 & \log n/4 \\
 \vdots & \vdots
 \end{array}$$



Big Omega (Ω) and Big Theta (Θ) Notations I

Big Omega (Ω)

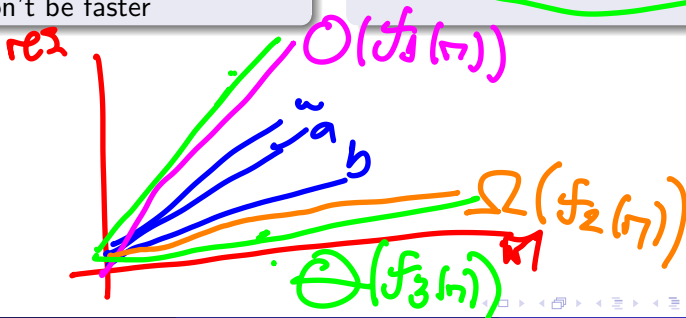
Lower Bound:

- Minimum time required
- Best-case analysis
- "At least this fast"
- Guarantees algorithm won't be faster

Big Theta (Θ)

Tight Bound:

- Exact growth rate
- Best and worst are same
- "Exactly this fast"
- Most precise description



Big Omega (Ω) and Big Theta (Θ) Notations II

Example

Merge Sort:

- Best case: $\Theta(n \log n)$
- Worst case: $O(n \log n)$
- Since best = worst: $\Theta(n \log n)$

Linear Search:

- Best case: $\Omega(1)$
- Worst case: $O(n)$
- No tight bound (best $\neq$ worst)



Study Case: Tower of Hanoi Problem I

Problem

Move n disks from **source** rod to **destination** rod using **auxiliary** rod. Rules: only move **one disk** at a time, never place **larger disk** on smaller one.

```
1: function Hanoi( $n$ , source, destination, auxiliary)
2: if  $n = 1$  then
3:   move disk from source to destination
4: else
5:   Hanoi( $n - 1$ , source, auxiliary, destination)
6:   move disk from source to destination
7:   Hanoi( $n - 1$ , auxiliary, destination, source)
8: end if
```



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```



Study Case: Tower of Hanoi Problem II

Recurrence Analysis

- $T(1) = 1$ (**base case**: move one disk)
- $T(n) = 2T(n - 1) + 1$ (**two recursive calls** + one move)

Solution:

- $T(n) = 2^n - 1$
- Complexity: $O(2^n)$ — **exponential!**

Real-World Implication

For 64 disks: $2^{64} - 1 \approx 18$ billion billion moves!



Study Case: Tower of Hanoi Problem II

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Solving Recurrence Relations

Common Solution Methods

- 1 **Substitution Method:** Guess and verify
- 2 **Recursion Tree:** Visualize recursive calls
- 3 **Master Theorem:** Direct formula for common cases
- 4 **Characteristic Equation:** For **homogeneous** recurrences



Solving Recurrence Relations

Example

Recursion Tree Method for $T(n) = 2T(n/2) + n$:

- Level 0: n work
- Level 1: $2 \times (n/2) = n$ work
- Level 2: $4 \times (n/4) = n$ work
- ...
- Total levels: $\log n$
- Total work: $n \times \log n = O(n \log n)$

Method Selection

- Simple cases: Direct solving
- Divide-and-conquer: Master theorem
- Complex cases: Substitution method

Solving Recurrence Relations

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Method Selection

- **Simple** cases: Direct solving
- **Divide-and-conquer**: Master theorem
- **Complex** cases: Substitution method

Master Theorem Introduction I

Master Theorem

For recurrences of the form $T(n) = aT(n/b) + f(n)$:

Case 1: $f(n)$ grows **slower** than $n^{(\log_b a)}$

- $T(n) = O(n^{(\log_b a)})$

Case 2: $f(n)$ grows at **same rate**

- $T(n) = O(n^{(\log_b a)} \log n)$

Case 3: $f(n)$ grows **faster**

- $T(n) = O(f(n))$



Master Theorem Introduction II

Examples

Merge Sort:

- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2: $O(n \log n)$

Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1, b = 2, f(n) = 1$
- Case 2: $O(\log n)$

When to Use

Master theorem works for many divide-and-conquer algorithms but has limitations.



Master Theorem Introduction II

Examples

Merge Sort:

- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2: $O(n \log n)$

Binary Search:

- $T(n) = T(n/2) + 1$
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When to Use

Master theorem works for many **divide-and-conquer** algorithms but has **limitations**.



Recursive Algorithm Complexity Calculation I

Systematic Approach

- 1 **Identify:** Base case(s) and recursive case(s)
- 2 **Set up:** Recurrence relation
- 3 **Solve:** Using appropriate method
- 4 **Verify:** Check solution makes sense



Recursive Algorithm Complexity Calculation II

Merge Sort Analysis

Step 1: Identify cases

- **Base:** Array size $1 \rightarrow O(1)$
- **Recursive:** Divide, sort halves, merge

Step 2: Set up recurrence

- $T(n) = 2T(n/2) + O(n)$

Step 3: Solve using Master Theorem

- Case 2: $T(n) = O(n \log n)$

Step 4: Verify

- Makes sense: **logarithmic levels**, **linear work** per level



Summary I

Key Takeaways

- **Complexity Analysis:** Essential for algorithm **comparison** and **selection**
- **Big O Notation:** Describes **growth rate** for large inputs
- **Recursion:** Powerful technique requiring careful **base case** design
- **Recurrence Relations:** Mathematical tools for **recursive algorithm** analysis



Summary II

Practical Skills Gained

- Count operations systematically
- Classify algorithms by growth rate
- Design recursive solutions correctly
- Analyze recursive algorithm complexity
- Apply Master Theorem to common patterns

Next Steps

- Practice complexity analysis on various algorithms
- Implement recursive solutions with proper optimization
- Study advanced recurrence solving techniques
- Apply knowledge to real-world algorithm selection



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Outline

- 1 Introduction to Complexity Analysis
- 2 Advanced Complexity Analysis



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

